

Natural Deduction

in Propositional Logic

Thanh-Sach LE

Faculty of Computer Science and Engineering
HoChiMinh City University of Technology

Dec. 20, 2012

- ① What is Natural Deduction?
- ② Direct Proof
 - using Rules of Implication
 - using Rules of Equivalence
- ③ Conditional Proof
- ④ Indirect Proof
- ⑤ Logical Truths

What is Natural Deduction?

Definition

Natural deduction is a method for establishing the validity of propositional type arguments that is both simpler and more enlightening than the method of truth tables.

Example

No.	English	Propositions	Notes
1.	Either Mai or Hoa breaks the cup.	$M \vee H$	Premise
2.	Mai does not break the cup.	$\neg M$	Premise
3.	So, Hoa breaks the cup.	H	Conclusion

What is Natural Deduction?

Deduction Problem

Input:

- Premises
- Conclusion

Requirement:

- Prove that the conclusion can be derived from given premises.
An assignment that makes the premises true also makes the conclusion true.

What is Natural Deduction?

HOW TO...

- A proof in natural deduction consists of a sequence of steps.
- Each step produces a proposition using a rule of inference.
There are 18 rules of inferences, including:
 - 8 rules of inference
 - 10 rules of equivalence
- The last steps yields the conclusion.

What is Natural Deduction?

Proof: Example

Premises:

- If Hoa is an outstanding student then Hoa has a good CV after graduation.
- If Hoa has a good CV after graduation then Hoa can find a job.
- Hoa can not find a job.

Premises:

- Hoa is not an outstanding student.

List of propositions

- ① H: Hoa is an outstanding student.
- ② G: Hoa has a good CV after graduation.
- ③ J: Hoa can find a job.

What is Natural Deduction?

Proof

Sequence of steps

No.	Propositions	Notes
1.	$H \Rightarrow G$	Premise
2.	$G \Rightarrow J$	Premise
3.	$\neg J$	Premise
	$\neg H$	Conclusion being proved
4.	$\neg G$	2, 3, Modus Tollens
5.	$\neg H$	1, 4, Modus Tollens
	□	proved

Propositional Language: Rules of inference

1. Modus ponens (MP):

$$\frac{p, p \Rightarrow q}{q}$$

2. Modus tollens (MT):

$$\frac{\neg q, p \Rightarrow q}{\neg p}$$

3. Hypothetical syllogism (HS):

$$\frac{p \Rightarrow q, q \Rightarrow r}{p \Rightarrow r}$$

4. Disjunctive syllogism (DS):

$$\frac{\neg q, p \vee q}{p} \quad (1)$$

Propositional Language: Rules of inference

5. Constructive dilemma (CD):

$$\frac{(p \Rightarrow q) \wedge (r \Rightarrow s), (p \vee r)}{q \vee s}$$

7. Conjunction (Conj):

$$\frac{p, q}{p \wedge q}$$

6. Simplification (Simp):

$$\frac{p \wedge q}{q}$$

8. Addition (Add):

$$\frac{p}{p \vee q} \quad (2)$$

Argument

If the Astros win the playoff, then the Braves will lose the pennant.
If the Astros do not win the playoff, then either Connolly or Davis will be fired. The Braves will not lose the pennant. Furthermore, Connolly will not be fired. Therefore, Davis will be fired.

Natural Deduction: Examples

Define Propositions

A : The Astros win the playoff.

B : The Braves will lose the pennant.

C : Connolly will be fired.

D : Davis will be fired.

Translate to propositional logic

1. $A \Rightarrow B$

2. $\neg A \Rightarrow (C \vee D)$

3. $\neg B$

4. $\neg C$

Conclusion: D

Natural Deduction: Examples

The conclusion is now derived from the premises via steps 5 through 7. The justification for each line is written to the immediate right:

Proof by Natural Deduction

No.	Proposition	Note
1.	$A \Rightarrow B$	Premise
2.	$\neg A \Rightarrow (C \vee D)$	Premise
3.	$\neg B$	Premise
4.	$\neg C$	Premise
5.	$A \Rightarrow B$	1,3, MT
6.	$C \vee D$	2,5,MP
7.	D	4,6,DS
	$\square$	

Natural Deduction: Strategies

- **Strategy 1:** Always begin by attempting to find the conclusion in the premises.

Natural Deduction: Strategies

- **Strategy 1:** Always begin by attempting to find the conclusion in the premises.
- **Strategy 2:** If the conclusion contains a letter that appears in the consequent of a conditional statement in the premises, consider obtaining that letter via modus ponens:
 1. $A \Rightarrow B$ Premise
 2. $C \vee A$ Premise
 3. A Premise
 B **Expected Conclusion**
 4. B 1, 3, MP
 $\square$

Natural Deduction: Strategies

- **Strategy 3:** If the conclusion contains a negated letter and that appears in the antecedent of a conditional statement in the premises, consider obtaining the negated letter via modus tollens:

1. $C \Rightarrow B$ Premise
 2. $A \rightarrow B$ Premise
 3. $\neg B$ Premise
 - $\neg A$ **Expected Conclusion**
 4. $\neg A$ 2, 3, MT
-

Natural Deduction: Strategies

- **Strategy 3:** If the conclusion contains a negated letter and that appears in the antecedent of a conditional statement in the premises, consider obtaining the negated letter via modus tollens:

1. $C \Rightarrow B$ Premise
 2. $A \rightarrow B$ Premise
 3. $\neg B$ Premise
 $\neg A$ **Expected Conclusion**
 4. $\neg A$ 2, 3, MT
-

- **Strategy 4:** If the conclusion is a conditional statement, consider obtaining it via hypothetical syllogism:

1. $B \Rightarrow C$ Premise
 2. $C \rightarrow A$ Premise
 3. $A \rightarrow B$ Premise
 $A \rightarrow C$ **Expected Conclusion**
 4. $A \rightarrow C$ 1, 3, HS
-

Natural Deduction: Strategies

- **Strategy 5:** If the conclusion contains a letter that appears in a disjunctive statement in the premises, consider obtaining that letter via disjunctive syllogism:

1. $A \Rightarrow B$ Premise
 2. $A \vee C$ Premise
 3. $\neg A$ Premise
 - C **Expected Conclusion**
 4. C 2, 3, DS
-

Natural Deduction: Strategies

- **Strategy 6:** If the conclusion contains a letter that appears in a conjunctive statement in the premises, consider obtaining that letter via simplification:

1. $A \Rightarrow B$ Premise
2. $C \wedge B$ Premise
3. $C \Rightarrow A$ Premise
- C **Expected Conclusion**
4. C 2, Simp
-

Natural Deduction: Strategies

- **Strategy 6:** If the conclusion contains a letter that appears in a conjunctive statement in the premises, consider obtaining that letter via simplification:

1. $A \Rightarrow B$ Premise
 2. $C \wedge B$ Premise
 3. $C \Rightarrow A$ Premise
 - C **Expected Conclusion**
 4. C 2, Simp
-

- **Strategy 7:** If the conclusion is a conjunctive statement, consider obtaining it via conjunction by first obtaining the individual conjuncts:

1. $A \Rightarrow C$ Premise
 2. B Premise
 3. $\neg C$ Premise
 - $B \wedge \neg C$ **Expected Conclusion**
 4. $B \wedge \neg C$ 2, 3, Conj
-

Natural Deduction: Strategies

- **Strategy 8:** If the conclusion is a disjunctive statement, consider obtaining it via constructive dilemma or addition:

1. $(A \Rightarrow B) \wedge (C \Rightarrow D)$ Premise

2. $B \wedge C$ Premise

3. $A \vee C$ Premise

$B \vee D$

Expected Conclusion

4. $B \vee D$ 1, 3, CD

□

Natural Deduction: Strategies

- **Strategy 8:** If the conclusion is a disjunctive statement, consider obtaining it via constructive dilemma or addition:

1.	$(A \Rightarrow B) \wedge (C \Rightarrow D)$	Premise
2.	$B \wedge C$	Premise
3.	$A \vee C$	Premise
	$B \vee D$	Expected Conclusion
4.	$B \vee D$	1, 3, CD
	$\square$	

- **Strategy 9:** If the conclusion contains a letter not found in the premises, addition must be used to obtain that letter:

1.	$A \vee C$	Premise
2.	B	Premise
3.	$C \Rightarrow D$	Premise
	$B \vee D$	Expected Conclusion
4.	$B \vee D$	2, Add
	$\square$	

Natural Deduction: Important Note

Rules of Implication

- To be applicable in natural deduction, certain lines in a proof must be interpreted as substitution instances of the argument form in question. Stated another way, the rules of implication are applicable only to whole lines in a proof.
- For example, step 3 in the following proof is not a legitimate application of modus ponens, because the first premise in the modus ponens rule is applied to only a part of line 1:
 1. $A \Rightarrow (B \Rightarrow C)$ Premise
 2. B Premise
 3. C 1,2, MP (**incorrect**)

Rules of Equivalence

- Logically equivalent statement forms can always replace one another in a proof sequence, the rules of replacement can be applied either to a whole line or to any part of a line.
- Step 2 in the following proof is a quite legitimate application of DeMorgans Rule, even though the rule is applied to only part of line 1:

1. $S \Rightarrow \neg(T \wedge U)$ Premise
2. $S \Rightarrow (\neg T \vee \neg U)$ 1, DM

Natural Deduction: Using rules of equivalence

Example

No.	Proposition	Note	Rules' Type
1.	$A \Rightarrow \neg(B \wedge C)$	Premise	Implication
2.	$A \wedge C$	Premise	Implication
.	$\neg B$	being proved	
3.	A	2, Simp	Implication
4.	$\neg(B \wedge C)$	1,3, MP	Implication
5.	$\neg B \vee \neg C$	4, DM	Equivalence: DeMorgan
6.	$C \wedge A$	2, Com	Implication
7.	C	6, Simp	Implication
8.	$\neg\neg C$	7, DN	Equivalence: Double Negation
9.	$\neg C \vee \neg B$	5, Com	Implication
10.	$\neg B$	8, 9, DS	Implication
	□	Proved	

Definition

Condition Proof Conditional proof is a method for obtaining a line in a proof sequence (either the conclusion or some intermediate line) that frequently offers the advantage of being both shorter and simpler to use than the direct method.

WHEN TO USE

In theory the method of conditional proof can be used to derive any line in a proof sequence, in practice it is usually reserved for obtaining lines that are expressed in the form of conditional statements.

HOW TO

The method consists of assuming the antecedent of the required conditional statement on one line, deriving the consequent on a subsequent line, and then **discharging** this sequence of lines in a conditional statement that exactly replicates the one that was to be obtained.

Condition Proof

Example

No.	Proposition	Note
1.	$A \Rightarrow (B \wedge C)$	Premise
2.	$(B \vee D) \Rightarrow E$	Premise
	$A \Rightarrow E$	being proved
3.	A	ACP
4.	$B \wedge C$	1, 3, MP
5.	B	4, Simp
6.	$B \vee D$	5, Add
7.	E	2,6, MP
8.	$A \Rightarrow E$	3-7, CP
	□	proved

ACP: Assumption for Conditional Proof

CP: Conditional Proof

Indirect Proof

Definition

Indirect Proof Indirect proof is a technique similar to conditional proof that can be used on any argument to derive either the conclusion or some intermediate line leading to the conclusion.

HOW TO

It consists of assuming the negation of the statement form to be obtained, using this assumption to derive a contradiction, and then concluding that the original assumption is false. This last step, of course, establishes the truth of the statement form to be obtained. The following proof sequence uses indirect proof to derive the conclusion:

Indirect Proof

Example

No.	Proposition	Note
1.	$(A \vee B) \Rightarrow (C \wedge D)$	Premise
2.	$C \Rightarrow \neg D$	Premise
	$\neg A$	being proved
3.	A	AIP
4.	$A \vee B$	3, Add
5.	$C \wedge D$	1, 4, MP
6.	C	5, Simp
7.	$\neg D$	2, 6, MP
8.	D	5, Com, Simp
9.	$D \wedge \neg D$	7, 8, Conj
10	$\neg A$)	3-9, IP
	$\square$	proved

AIP: Assumption for Indirect Proof

CP: Indirect Proof